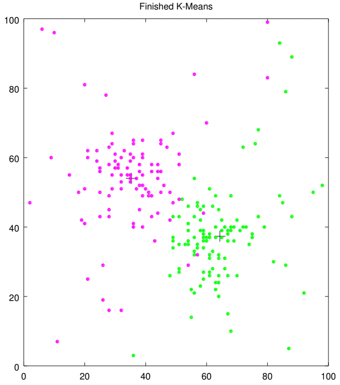
Wearable Technologies and Applications (Wearable Informatics)

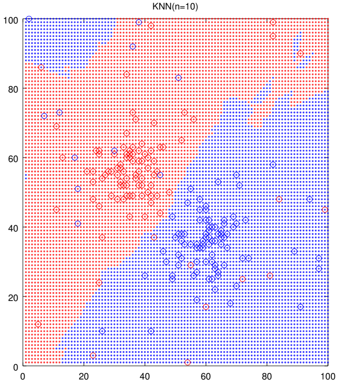
Winfree

Lecture 6

Comparing K-Means and KNN

Neither of these are great for time-series analysis.





K-Means pulls data apart, after you have the data.

Use the labeled data, can be used as a live model.

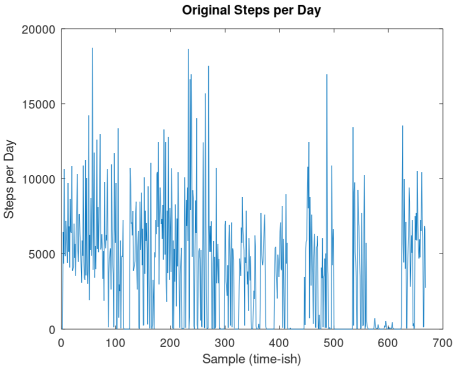
Steps per Day

Suppose we have time series data

- Steps per day
- Steps per hour
- Heart rate per minute

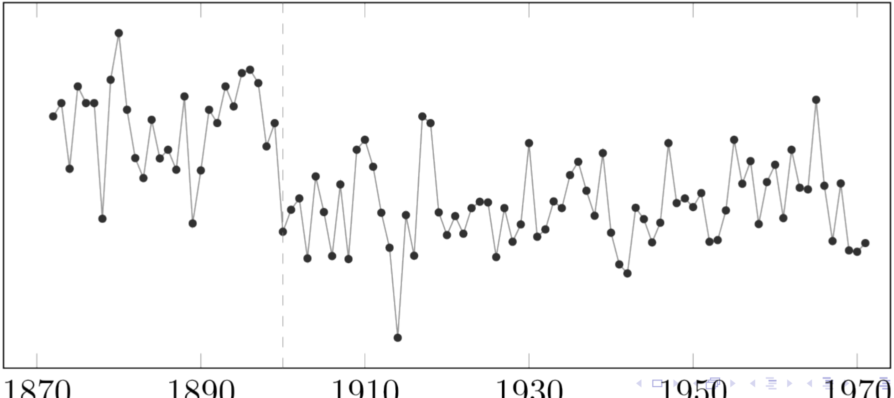
You already looked at this in HW2, right?

How did you approach that there?

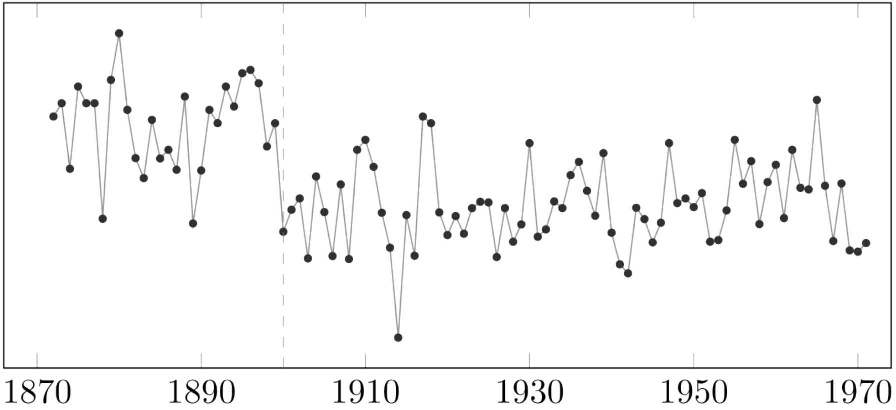


Overview

In statistical analysis, change detection or change point detection tries to identify times when the probability distribution of a stochastic process or time series changes. In general the problem concerns **both** detecting *whether or not* a change has occurred, or whether several changes might have occurred, and *identifying the times of any such changes*.



Overview



Yearly volume of the Nile river at Aswan, an example of time series data commonly used in change detection. Dotted line denotes a detected change point when Old Aswan Dam was built in 1902.

Algorithm

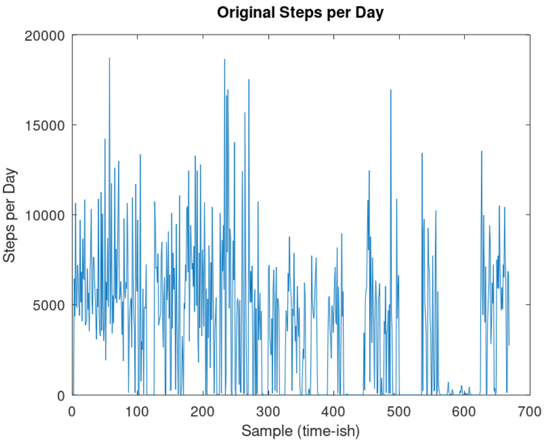
$$\vec{t} = \text{set of n timestamps}$$
$$\vec{x} = \text{set of n measures}$$
$$\bar{x} = \text{mean of measures}$$
$$\vec{\varepsilon} = x - \bar{x} = \text{errors from mean}$$
$$y_i = \sum_{j=i}^{n} \varepsilon_j$$
$$\vec{y} = \text{set of cumulative sums}$$
$$R = max(\vec{y}) - min(\vec{y})$$
$$= \text{range}$$

1. Establish original cumulative sum ($\vec{y}$)
2. Find / save $R_{original}$
3. Find / save i of $max(|\vec{y}|)$
4. Bootstrap many times ($K = 10 \ldots 1000$)
   4.1 Randomly re-order x
   4.2 Find / save new $\vec{y}$
5. Find / save $R_{bootstraps}$
6. Confidence $= \frac{\sum(R_{bootstraps} < R_{original})}{K}$
7. If Confidence $>$ some threshold, then i of $max(|\vec{y}|)$ is CP

Example

How about that data again?

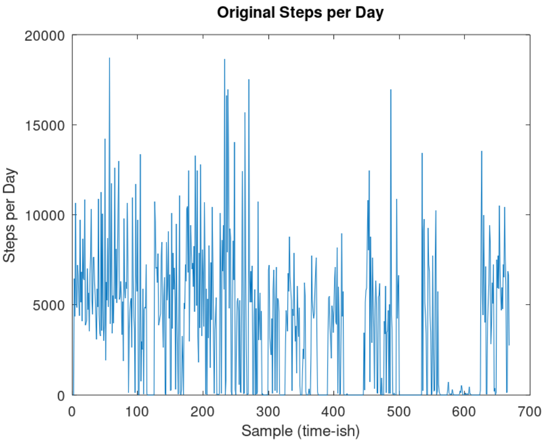
Let's ignore timestamps for this example.



Validate / Clean Up

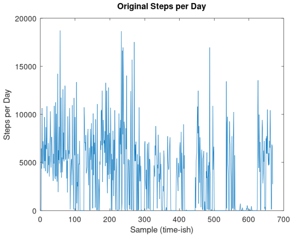
Notice how we have some zero or near zero measures?

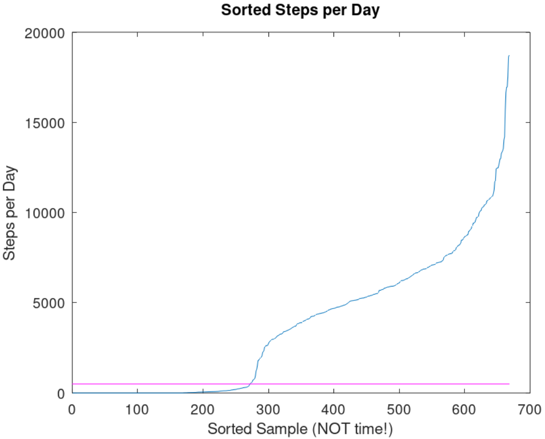
Let's assume those are times when the Fitbit wasn't being worn.



Validate / Clean Up

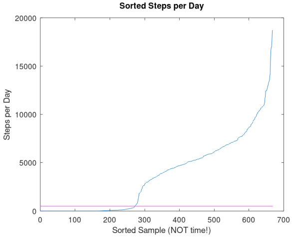
Notice how we have some zero or near zero measures?
Let's assume those are times when the Fitbit wasn't being worn.
We can sort these and plot, without regard for the timestamp.

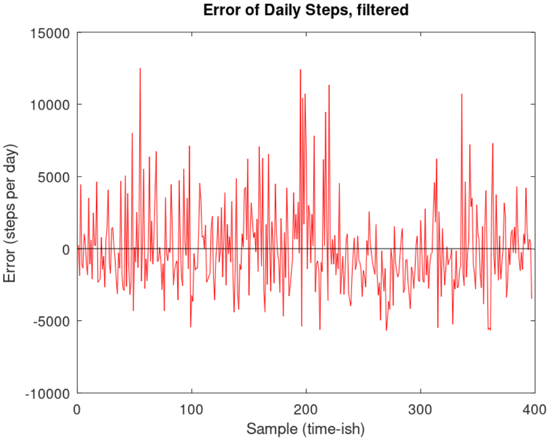




Subtract off the Mean

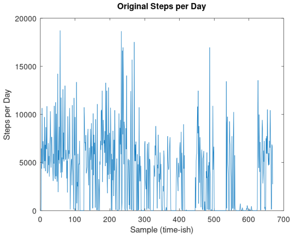
With the “off” days out of the way, we can subtract the mean from all points.

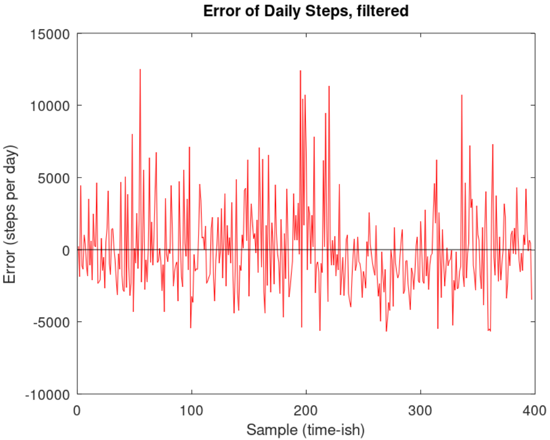




Subtract off the Mean

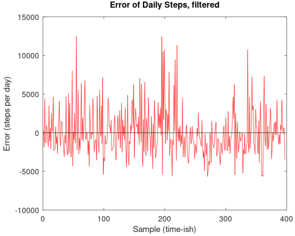
With the "off" days out of the way, we can subtract the mean from all points.
Looks pretty similar, but ...

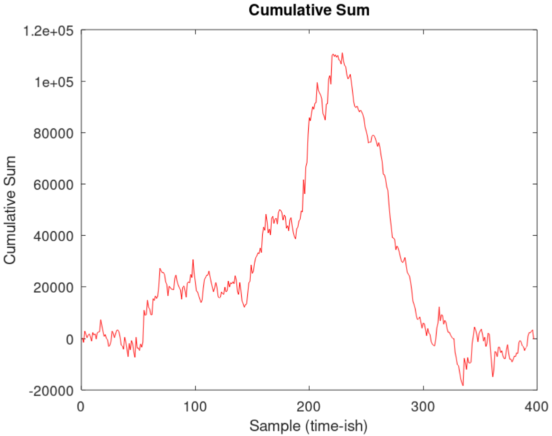




Subtract off the Mean

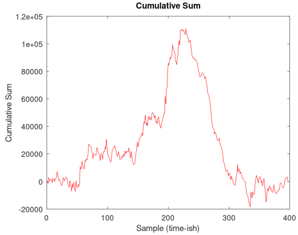
Calculate the cumulative sum of the errors, in the original data. See how this starts at zero and returns to zero?

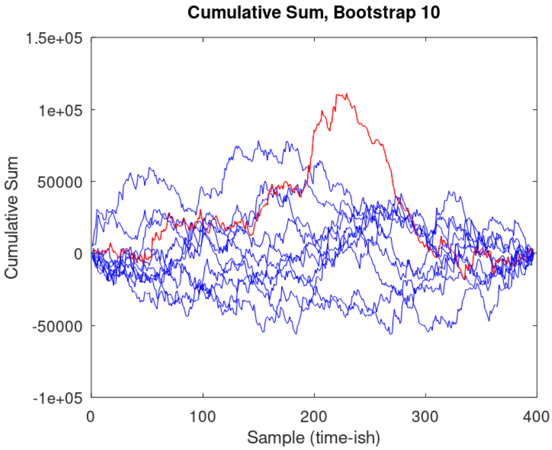




Bootstrap it

Randomize the order of the errors (or x and then recalc ε).
Calculate the cumulative sum of the errors.
Do this "many" times.



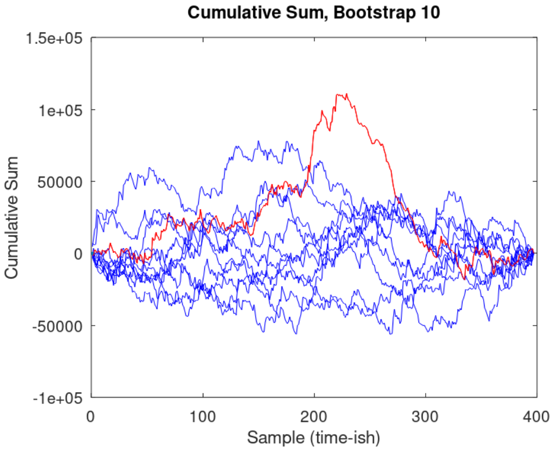


Confidence

Statistically, how many times was the range of the original data larger than the bootstrapped reorders?

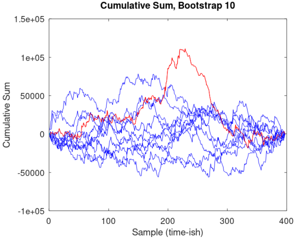
Conf. $= \frac{\sum(R_{bootstraps} < R_{original})}{K}$

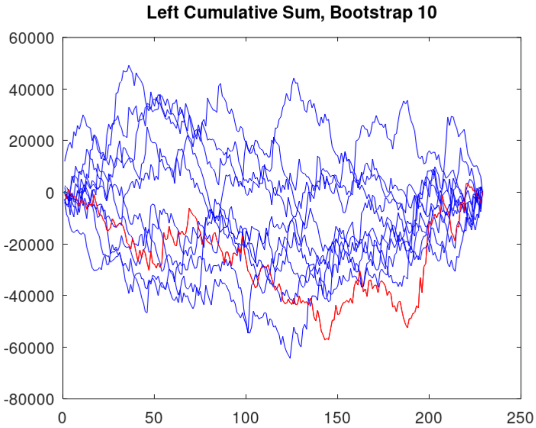
If Conf. $>$ some threshold, then i of $max(|\vec{y}|)$ is change point!



Rinse and Repeat

Because we found a change point, we need to cut the data in half and repeat this process on the left, and the right sections of data.





10 vs 1000

Best practice is to bootstrap some large number of times. 1000 is generally considered sufficient. 10, from our example, is *not*.